12.7 Nutritional implications of cancer

There are three major nutrition related problems that are often exhibited in a cancer patient:

1. Anorexia
2. Drastic weight loss
3. Cancer cachexia.

1. Anorexia

Anorexia means "loss of appetite". The anorexia of cancer may be related to depression, the trauma of the disease, as well as to the side effects of treatment. Chronic anorexia observed in cancer leads to "cancer cachexia which is characterised by severe loss of muscle mass and body weight.

2. Drastic weight loss

Weight loss in cancer patients may be attributed to a chronic loss of appetite, persistent fever and changes in taste perception that reduces food intake. Individuals with cancer often lose weight drastically. Cancer cells compete with normal cells in utilizing nutrients and energy reserves of the body to grow. On the other hand, normal cells require energy to fight this abnormal growth. This increase in demand for nutrients and energy by both normal and cancer cells, results in a hypermetabolic state that causes a significant loss of weight. As cancer progresses, the individual becomes weaker and is less able to fight infections.

3. Cancer cachexia

Cancer cachexia is a "wasting syndrome" characterized by extreme loss of the body's muscle and fat mass, anorexia, asthenia (abnormal physical weakness) and anemia (a deficiency of red cells or of haemoglobin in the blood). Tumor cells release substances that reduce appetite. Cancer and its treatment can also cause severe nausea and may damage the digestive system, leading to poor absorption of nutrients. As the body gets fewer nutrients, it burns fat and muscle. Cancer cachexia is the most common indicator of advanced stage of the disease.



Fig 12.5 Wasting syndrome – Cancer cachexia

Consequences of malnutrition in cancer are:

- Severe loss of tissue proteins
- Compromised immune function increasing the individual's susceptibility to infections.
- Poor wound healing
- Abnormal reactions to cancer drugs and treatment

Activity : 1

Match the following

Anorexia	-	wasting syndrome
Chemotherapy	-	spread of cancer to other parts of the body
Cachexia	-	The study of cancer
Oncology	-	treatment procedure for cancer
Metastasis	-	loss of appetite

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